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TECH STACK

BACKEND
Python 3.13
FastAPI, Flask
Pydantic v2
SQLAlchemy 2 (async)
Alembic, httpx
REST API

ARCHITECTURE
Microservices
API Gateway
Temporal (workflows)
Webhooks, idempotency
Layers: API / Service / Repository

DATA
PostgreSQL
MySQL, MongoDB
Redis

AI / LLM
OpenAI (function calling)
Claude, Ollama
Prompt engineering

DEVOPS / CLOUD
Docker, Compose
Kubernetes, Helm, k3s
Traefik, Nginx
cert-manager, Let's Encrypt
GitHub Actions, GHCR
Linux (VPS, ufw, SSH)

TESTING / QUALITY
pytest, testcontainers
respx, pytest-cov
mypy, ruff
pre-commit, gitleaks
Conventional Commits

INTEGRATIONS
Stripe (Checkout, webhooks)
Auth0 (OIDC, JWT)
Brevo SMTP (DKIM)
TMDB, Fakturownia

FRONTEND
React (basics)
Vite

DAMIAN KOWALCZYK

PYTHON BACKEND DEVELOPER · MICROSERVICES & AI

PROFILE

I am a Python Backend Developer. I independently designed, tested and **deployed to production** a microservice system for booking cinema tickets — from the first line of code to a public demo on **Kubernetes** with HTTPS and automated deployments. I build asynchronous APIs in **FastAPI**, business processes as durable **Temporal** workflows, **Stripe** payment integrations and **AI assistants with function calling**. I focus on clean architecture, strict typing (mypy) and **100% test coverage**. I also bring 13 years of running my own business — I understand business processes, customers and ownership of results.

6	100%	Live	13 yrs
APPS IN A KUBERNETES CLUSTER	TEST COVERAGE ACROSS 5 SERVICES	PRODUCTION DEMO WITH CI/CD	RUNNING MY OWN BUSINESS

PROJECTS

Cinema Reservation System

Personal project · 2026 · cinema.damian-kowalczyk.dev · GitHub

- Designed and built a ticket booking platform from scratch: **5 FastAPI services** (gateway, cinema, payment, AI, worker) + React frontend, a separate PostgreSQL database per service
- Reservation lifecycle as a **durable Temporal workflow** — seat hold with expiry, payment signal and ticket delivery with an **HMAC-signed QR code**; resilient to restarts and redeploys
- **Stripe Checkout** payments: webhook signature verification, idempotent sessions and an automatic refund when the payment arrives after the reservation has expired
- **AI assistant** (OpenAI function calling) searches screenings, shows the seat map and creates a reservation from a conversation; round and token limits, chat history in Redis with TTL
- Authentication with **Auth0 (OIDC/JWT) in the API Gateway only**, admin/user roles, token in an HttpOnly cookie, rate limiting
- Deployment on **k3s (VPS)**: one generic **Helm chart** for 6 apps, migrations in an initContainer, **HTTPS (cert-manager, Let's Encrypt)**, e-mails via Brevo with DKIM/DMARC
- **CI/CD in GitHub Actions**: lint and secret scanning, matrix tests, images in GHCR, automatic **helm upgrade** on every push
- **100% test coverage** (pytest, testcontainers with real PostgreSQL, respx), mypy

▸ Python, FastAPI, SQLAlchemy 2, Alembic, PostgreSQL, Redis, Temporal, Stripe, Auth0, OpenAI, React, Docker, Kubernetes, Helm, GitHub Actions

KEY STRENGTHS

- Project from scratch to production, independently
- Microservices and asynchronous communication
- Durable business processes (Temporal)
- Payment integrations and webhooks
- AI assistants with function calling
- Testing, typing, CI/CD
- Kubernetes and VPS deployments

SOFT SKILLS

✓ **Business mindset**
13 years of my own business — sales, customers, processes.

✓ **Independence**
Entire project: architecture, code, tests, deployment.

✓ **Customer communication**
Turning business needs into technical requirements.

✓ **Fast learner**
Temporal, Kubernetes and Stripe learned hands-on, in a real project.

LANGUAGES

Polish — native

English — B2

EDUCATION

Wyższa Szkoła Kształcenia Zawodowego

Software Engineering, BEng studies

10.2025 – present

KM-Programs

Mentoring program: Python Back-end Developer

09.2024 – 05.2026

WORK EXPERIENCE

Python Backend Developer — Internship

Devs-Mentoring · remote · 05.2026 – present

- Developing a microservice system (FastAPI, PostgreSQL, Temporal) under the supervision of a Senior Developer — every change goes through a **Pull Request and code review**
- Applying review feedback: correct ordering of transactions and workflows, error handling, unit and integration tests
- Containerization and deployment: **Docker, Helm, Kubernetes (k3s)**, CI/CD pipeline in GitHub Actions
- Following project standards: Conventional Commits, pre-commit (ruff, mypy, gitleaks)

Business Owner — automotive / e-commerce

Sole proprietorship · 05.2012 – 06.2025

- Ran sales and e-commerce operations for hundreds of customers per year
- Optimized business processes that directly affected the company's results
- Direct customer contact and turning customer needs into concrete solutions

PROJECTS (CONTINUED)

Auth Microservices System · personal project

- Authentication with an **API Gateway** and an isolated user service (FastAPI, async, httpx)
- **JWT (access + refresh)** in HttpOnly cookies, **MFA (TOTP)** and **RBAC** access control

▸ Python, FastAPI, MongoDB (Beanie), JWT, TOTP, Docker Compose

Course Platform Microservice · personal project

- E-learning backend in **Flask**: Pydantic validation, JWT + MFA, background jobs (APScheduler)
- Hybrid data model: **MySQL** (SQLAlchemy, Alembic) + **MongoDB**; Docker and Nginx

▸ Python, Flask, MySQL, MongoDB, SQLAlchemy, Alembic, Docker, Nginx

AI Web Summarizer · personal project

- CLI: extracts text from web pages and generates **LLM summaries** (OpenAI / Ollama), YAML prompts, export to multiple formats

▸ Python, OpenAI API, Ollama, BeautifulSoup, pytest, mypy

Insurance Client Manager · personal project

- Client data aggregation (pandas), Excel reports, e-mail notifications and **Fakturownia API** integration

▸ Python, pandas, openpyxl, REST API, pytest, mypy

I HEREBY CONSENT TO THE PROCESSING OF MY PERSONAL DATA FOR THE PURPOSES OF THE RECRUITMENT PROCESS (IN ACCORDANCE WITH REGULATION (EU) 2016/679 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL OF 27 APRIL 2016 — GDPR).